

Onatola Timilehin Faruq

Web3 Infrastructure Engineer | SDK Development | Full-Stack dApps

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Summary

Full-stack Web3 engineer specializing in middleware, SDK development, and developer tooling. Built OmniSwap SDK, a cross-chain swap library with chain adapters for 6 blockchains and comprehensive documentation. Ships production dApps using React, Next.js, TypeScript, Viem, Wagmi, and Ethers.js. Experience building on BNB Chain (TruthBounty, 2nd place Seedify Hackathon).

Technical Expertise

Frontend: React | Next.js 14 | TypeScript | Tailwind CSS | Three.js | React Three Fiber
Web3 Integration: Viem | Wagmi | Ethers.js v6 | Web3.js | Wallet Connectors (MetaMask, WalletConnect, HashPack, Blade)
SDK & Middleware: API Design | Chain Adapters | Error Handling | Retry Logic | TypeScript SDKs | Developer Documentation
Smart Contracts: Solidity (Advanced) | Rust/Anchor (Intermediate) | Foundry | Hardhat
Backend: Node.js | Express | PostgreSQL | Redis | WebSocket | GraphQL | REST APIs
Infrastructure: The Graph (Subgraphs) | IPFS/Arweave | Docker | GitHub Actions CI/CD | Tenderly
Chains: Ethereum | BNB Chain | Base | Arbitrum | Optimism | Solana | Hedera

Professional Experience

Blockchain Engineer | DeFiConnectCredit | Jan 2025 - Present

- Architected modular frontend components for multi-protocol DeFi dashboard integrating Aave v3, Uniswap v3/v4, Curve, and GMX
- Built reusable React hooks and TypeScript utilities for wallet connection, transaction signing, and contract interactions across 4 DeFi protocols
- Reduced transaction costs by **35%** through gas optimization, improving UX for end users
- Created deployment automation suite cutting deployment time from 2 hours to 15 minutes
- Established Foundry testing framework with **95%+ code coverage** across integration suite

Blockchain Engineer (Intern) | DeFiConnectCredit | Jun 2024 - Dec 2024

- Contributed to lending protocol frontend with collateral management and liquidation mechanism interfaces
- Conducted security reviews on **15+ contracts** pre-deployment
- Built ERC721 incentive system with Merkle tree proofs serving 500+ users

SDK & Middleware Projects

OmniSwap SDK — Cross-Chain Swap Middleware

Production-grade TypeScript SDK enabling atomic swaps across 6 heterogeneous blockchains

- Architected chain-specific adapter pattern: unified interface, per-chain implementation (Zcash, Osmosis, Mina, Aztec, Miden, Fhenix)
- Implemented HTLC-based swap protocol with automated refund management and timeout handling
- Built typed error system with exponential backoff retry logic for RPC resilience
- Created **70KB+ documentation**: API reference, step-by-step tutorials, 5 working examples, migration guides
- Set up CI/CD pipeline with GitHub Actions for automated testing and publishing

TypeScript Multi-Chain SDK Design Developer Tooling

OmniShield Wallet — Multi-Chain HD Wallet

Developer-facing wallet SDK supporting 6 blockchains with unified interface

- Implemented BIP-39/32/44 key derivation across multiple chain types
- Built chain-specific adapters with unified API for balance queries and transactions
- Created both React web interface and React Native (Expo) mobile app

TypeScript React React Native Wallet Integration

Full-Stack dApp Projects

Comic Pad — NFT Marketplace (Production)

Full-stack platform on Hedera with complex financial UX and multi-wallet support

- Solo-built complete marketplace: smart contracts, backend API, React/Next.js frontend
- Integrated **3 wallet providers** (HashPack, Blade, MetaMask) with seamless connection flow
- Built real-time WebSocket server delivering updates to **1K+ concurrent users** (<50ms latency)
- Designed UX for Dutch/English auctions, royalty distribution, dynamic pricing, and token-gated content

Next.js TypeScript PostgreSQL Redis WebSocket Hedera

TruthBounty — Prediction Market Protocol (BNB Chain)

2nd Place, Seedify Hackathon | Reputation system integrating 13 prediction market platforms

- Built data aggregation layer across PancakeSwap Prediction, Polymarket, Azuro, Thales, and 10 others
- Designed soulbound NFT credentials with on-chain SVG generation
- Created custom subgraphs for cross-chain indexing and real-time data fetching
- Built Next.js dashboard presenting complex reputation data through clean, scannable UI

Solidity Next.js The Graph BNB Chain Soulbound NFTs

Sentiment — Uniswap v4 Dynamic Fee Hook

UHI7 Incubator Capstone | Dynamic fee hook consuming data from 8 off-chain sources

- Built data pipeline integrating multiple oracle sources with staleness detection via Chainlink Automation and Gelato
- Achieved **<30k gas overhead** per swap through optimization
- Graduated from Uniswap Hook Incubator (15% acceptance rate)

Solidity Foundry Chainlink Uniswap v4

Privacy Chronicles — Interactive Education Platform

Content Creation Track Winner, Zyperpunk Hackathon (Zcash & Mina Foundations)

- Built immersive 3D solar system in React Three Fiber with interactive episode navigation
- Implemented accessibility-first design with voice narration for 70+ page comic series

Next.js Three.js React Three Fiber TypeScript

Certifications & Training

- **Uniswap Hook Incubator (UHI7)** — Graduate (15% acceptance rate)
- **Cyfrin Updraft** — DeFi Security & Protocol Engineering (Completed)
- **School of Solana** — Ackee Blockchain (Rust/Anchor Development)
- **Hashgraph Developer Certification** — The Hashgraph Association

Education

BSc Physiotherapy | University of Ibadan, College of Medicine | 2024

Portfolio: [faruukku.vercel.app](#) | GitHub: [github.com/Leihyn](#) | References available upon request